SMU SINGAPORE MANAGEMENT UNIVERSITY | School of Computing and Information Systems

CS425: Generative AI for Natural Language Communication

Group Project Proposal · AY2026/27 Term 1

EmoDial

Emotion-Aware Empathetic Dialogue with Emotion Transition Gating

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Main text: pages 1–3, including references (within the 3-page limit).
Appendices A–D are supplementary detail; the main text is complete without them.

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1 Motivation and Objectives

Models trained on EmpatheticDialogues (ED) come across as more empathetic than general chit-chat models (Rashkin et al., 2019). Emotion-conditioned generators such as ECM (Zhou et al., 2018) and MoEL (Lin et al., 2019) steer the reply using the emotion the user is expressing at that moment. The problem is that people's emotions change during a conversation, and even emotion recognisers struggle with these shifts (Ghosal et al., 2020; Agarwal et al., 2021). Take a user who starts off anxious because their dream university rejected them, and then tells us they have won a scholarship somewhere else. A good listener picks up on the relief and pride, but a model that only sees one emotion label can miss this and keep responding to the earlier, negative mood.

Our project, **EmoDial**, tries to address this by adding a small **Emotion Transition Gate (ETG)** on top of a fine-tuned DialoGPT. The gate looks at how the user's predicted emotion distribution has changed since their last turn, and learns for each turn how much weight to give that change compared with the current emotion. **Our hypothesis** is that ETG will produce replies that people rate as more empathetic, with less unwarranted negative echoing than static emotion conditioning, without hurting fluency. We break this into three research questions:

- **RQ1.** Does giving the generator the recognised emotion (B1) make its replies more empathetic than having no emotion input at all (B0)?
- **RQ2.** Does transition-aware gating (ETG) improve empathy and relevance over static conditioning? If so, is the gain really coming from the shift signal and the gate rather than from extra parameters? We test this with two ablations, ETG-noΔ and ETG-noGate.
- **RQ3.** When the user's emotion shifts, does ETG give fewer *negative-echo* replies (replies that carry negative emotion themselves), compared with how often the human listeners did?

2 Related Work and Contribution

In emotion recognition, DialogueRNN (Majumder et al., 2019) and COSMIC (Ghosal et al., 2020) track each speaker's state across turns, and Agarwal et al. (2021) and Kumar et al. (2022) look specifically at emotion shifts and flips. These papers recognise or explain shifts, but do not use them to generate replies. For generation, ECM and MoEL condition on a single current emotion, while prompt and prefix tuning (Lester et al., 2021; Li & Liang, 2021) show that a language model can be steered with learned continuous vectors. The two papers closest to ours are **Emp-RFT** (Kim et al., 2022), which uses a dedicated encoder to model how emotion, keywords and meaning change between utterances, and **EPDC** (Mao et al., 2025), which measures a VADER sentiment shift between the user's *first* and *current* utterance, buckets it into positive, negative or neutral, and feeds it in through attention.

How EmoDial is different. First, our Δ_t is the difference between two full 7-class emotion distributions, so it shows *which* emotions changed, not just whether sentiment went up or down. Second, we compare *consecutive* user turns, so the model reacts to the latest change instead of the overall drift since the start. Third, a learned gate g_t decides on each turn how much the shift should matter, and since it is a single number we can check whether it actually rises with $\|\Delta_t\|_1$. Finally, ETG is a small add-on to a standard pretrained model, tested with ablations that isolate Δ_t and the gate.

3 Data

- **EmpatheticDialogues (ED)** (Rashkin et al., 2019) is our main generation dataset: 24,850 speaker–listener conversations, split 19,533 / 2,770 / 2,547. Conversations average about 4.3 utterances, so most have only two speaker turns and at most one shift. ED also has just one emotion label per conversation (for the situation), so it cannot train a turn-level recogniser.
- **ESConv** (Liu et al., 2021) has 1,300 help-seeker–supporter conversations averaging about 30 utterances, and the seeker's mood usually changes as they are supported. This gives us the longer, shift-heavy conversations ETG is designed for. We train on the ED and ESConv training sets together and report each test set separately.
- **For the recogniser** we use DailyDialog (Li et al., 2017; 13,118 dialogues) and MELD (Poria et al., 2019; 1,433 dialogues, about 13k utterances, text only). Both label every utterance with the same 7 emotions (anger, disgust, fear, joy, sadness, surprise, neutral), so K = 7 with no label mapping. We picked MELD over IEMOCAP because IEMOCAP needs a licence request. Most utterances in both are neutral, so we use a class-weighted loss and report per-class F1.
- **Preprocessing.** For every listener or supporter reply, the context is the last 3 turns. The frozen recogniser gives p_t for each user turn, and $\Delta_t = p_t - p_{t-1}$ (0 for the first user turn); both are cached. By **12 Oct** we will

report the Δ distribution and the number of shift turns ($\|\Delta_t\|_1 > \tau$) per dataset, which sets the RQ3 and human-evaluation sample sizes.

4 Architecture and Method

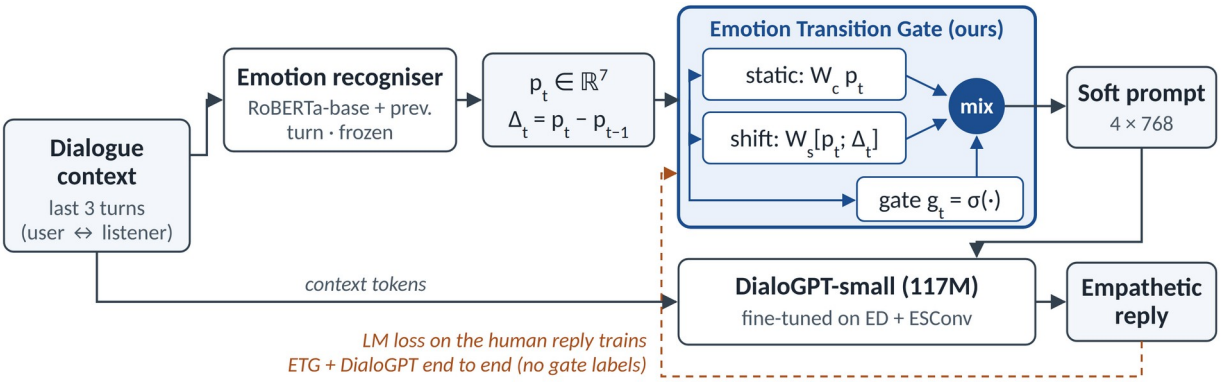


Figure 1. EmoDial pipeline. The shaded block (ETG) is our contribution; only the recogniser is frozen.

Emotion recogniser. We fine-tune RoBERTa-base (Liu et al., 2019) as a classifier that reads the current user utterance together with the previous turn, train it on DailyDialog + MELD, and then freeze it. We chose this over DialogueRNN because it is simpler to build and runs directly on ED and ESConv text; DialogueRNN stays as a stretch comparison. **Generator.** DialoGPT-small (Zhang et al., 2020). **ETG** turns (p_t, Δ_t) into m = 4 soft-prompt vectors (4 × 768) that go in front of the context embeddings. In standard prompt tuning the prompt is one fixed set of learned vectors; ours is recomputed every turn from the user's emotion state:

$$g_t = \sigma(\, w_g^{\top}\, [\, \|\Delta_t\|_1 \,;\, p_t \,] + b_g \,) \tag{1}$$

$$s_t = (1 - g_t) \cdot W_c\, p_t \;+\; g_t \cdot W_s\, [\, p_t \,;\, \Delta_t \,], \quad s_t \in \mathbb{R}^{4\times 768} \tag{2}$$

ETG and DialoGPT are trained together with the usual language-modelling loss on the human reply, so the gate needs no labels of its own. To keep things fair, we first fine-tune DialoGPT on ED + ESConv, and every system (B0 included) then continues from that same checkpoint for the same number of steps, with or without its emotion prompt. Any difference between systems therefore comes from the emotion input. All systems decode the same way, with beam search (width 4, no repeated trigrams):

System	Emotion input to the generator	Role
B0	none (plain fine-tuned DialoGPT)	emotion-blind baseline
B1	static prompt $W_c\, p_t$ only (g = 0)	current-emotion conditioning
ETG-noΔ	full ETG with Δ_t fixed to 0	controls for extra parameters
ETG-noGate	g_t fixed to 1 (shift branch only)	tests the learned gate
ETG	full model, Eqs. (1)–(2)	proposed

Everything is built in PyTorch and Hugging Face on Colab (T4) or the SCIS GPU, and no LLM APIs are used. With a 117M-parameter generator, cached p_t/Δ_t and a shared B0 checkpoint, each variant should take only a few GPU-hours. Appendix A walks through a worked example and Appendix B lists our planned settings.

5 Evaluation

Question	Comparison	Evidence
Recogniser	on DailyDialog/MELD test	weighted and per-class F1
RQ1	B1 vs. B0	human empathy; perplexity, Distinct-1/2
RQ2	ETG vs. B1, ETG-noΔ, ETG-noGate	human empathy and relevance; pairwise ETG vs. B1; emotion agreement with the human reply; g_t vs. $\|\Delta_t\|_1$
RQ3	all systems on the shift subset	negative-echo rate vs. human-reply rate
Prompt use	swap in another context's p_t, Δ_t	change in reply emotion and perplexity

Human evaluation. Raters see replies without system names and in random order. (a) For 20 test contexts, half of them shift turns, at least three raters score replies from B0, B1, ETG-noΔ and ETG from 1 to 5 on *empathy*, *relevance* and *fluency*, and we report Krippendorff's α. These also form our report's qualitative analysis. (b) For 50 shift-turn contexts, raters pick the better reply from ETG and B1 (sign test), which is more sensitive than 1–5 scores. Raters are group members, so blinding matters. Automatic metrics are averaged over 2 seeds; the rating guide is in Appendix C. **Negative echo.** A reply counts as negative if its combined anger, disgust, fear and sadness probability is above 0.5 under a *separate* classifier (Hartmann, 2022). We do not use our own recogniser for this, as that would be circular. The same classifier also gives us *emotion*

agreement between generated and human replies. We only claim that ETG reduces echoing, not that it calms the user down.

6 Plan, Risks and Team

5–16 Oct: data pipeline, recogniser, Δ statistics (12 Oct) and B0. **19–30 Oct:** B1, ETG, ETG-noΔ and ETG-noGate, plus tuning. **2–6 Nov:** automatic and human evaluation, gate and error analysis, so that at least 90% of results are ready for the **10 Nov** presentation. **11–27 Nov:** remaining analysis, then the report and code (27 Nov).

Risk	Mitigation
Recogniser error / domain gap	report F1; check ED predictions vs. grouped ED labels; all variants share p_t
Few shifts in ED	ESConv adds long, shift-rich dialogues; Δ checkpoint 12 Oct
ETG shows no gain	report as a negative result; ablations and gate analysis still answer RQ2

Role	Proposed duties	Member
Recogniser	DailyDialog + MELD prep, RoBERTa classifier, per-class F1, ED check	Denzil
Data pipeline	ED + ESConv contexts, p_t/Δ_t caching, Δ statistics, B0 baseline	Stevson
Generator & ETG	soft prompts, gate, training the four emotion variants, tuning	York Khang
Auto evaluation	perplexity, Distinct-n, negative echo, emotion agreement, swap test	Mikolaj
Human eval & writing	rating protocol, α, error analysis; leads report and slides	Aleksandra

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Appendices (supplementary; outside the 3-page main text)

Appendix A. Worked example of ETG (illustrative values, not results)

Recogniser classes, in order: anger, disgust, fear, joy, neutral, sadness, surprise. In ED terms the user is the *speaker*.

Turn	Who	Utterance	Recogniser output p_t (top classes)
1	User	"I didn't get into my dream university. I've been so anxious about what comes next."	fear 0.45, sadness 0.40, neutral 0.08
2	Listener	"I'm sorry, that's really tough. Do you have other options?"	(listener turns are not scored)
3	User	"Actually, I just found out I got a full scholarship at another school!"	joy 0.70, surprise 0.15, neutral 0.06

Shift. $\Delta_3 = p_3 - p_1$ gives joy +0.68, fear −0.42, sadness −0.35 and surprise +0.13 (the rest are close to 0), so $\|\Delta_3\|_1 = 1.62$, which is a big shift. We would expect the gate to be high here (around 0.9), so the shift branch $W_s[p_3; \Delta_3]$ makes up most of the soft prompt. When nothing changes between turns ($\Delta \approx 0$), g should drop and ETG should behave much like B1. We will check whether this really happens by plotting g_t against $\|\Delta_t\|_1$ on the test set.

System	What it is told	Typical reply we expect (hypothetical)
B0	text only	"I'm sorry to hear that. What will you do next?" (anchored on the rejection: negative echo)
B1	current emotion: joy	"That's great, congratulations!" (right emotion, misses the turnaround)
ETG	joy, plus fear/sadness → joy	"What a turnaround! After the rejection that must be such a relief. Congratulations!"

Appendix B. Implementation details and planned settings

These are starting points. We will tune them on the validation splits and report the final values in our report.

Component	Planned setting
Recogniser	RoBERTa-base; input "previous turn </s></s> current utterance", max 128 tokens; AdamW, lr 2e-5, batch 32, up to 5 epochs with early stopping on validation weighted F1; class-weighted cross-entropy
Generator input	last 3 turns joined by the EOS token, max 256 tokens; target reply truncated to 64 tokens
Optimisation	AdamW, lr 5e-5 for DialoGPT and 1e-3 for ETG; effective batch 16 via gradient accumulation; fp16
Training schedule	B0: 3 epochs on ED + ESConv; every system then trains 1 more epoch from the B0 checkpoint; best checkpoint by validation perplexity; 2 seeds
ETG size	W_c: 7 → 3,072; W_s: 14 → 3,072; gate: 8 → 1; about 70.7k parameters (0.06% of the generator)
Shift threshold τ	chosen on validation (e.g. top quartile of non-zero $\|\Delta_t\|_1$) and fixed before testing
Decoding	beam 4, no repeated trigrams, at most 40 new tokens; identical for all systems
Judge metrics	Hartmann (2022) classifier: negative mass = anger + disgust + fear + sadness > 0.5; emotion agreement = top-1 match and Jensen–Shannon distance to the human reply
Compute estimate	Colab T4 (16 GB) or SCIS GPU: recogniser < 1 h, B0 about 2–3 h, each variant about 1 h per seed

Appendix C. Human evaluation guide

Score	Empathy	Relevance	Fluency
1	ignores or misreads how the user feels	off-topic or contradicts the context	ungrammatical or incoherent
3	generic acknowledgement ("that's nice")	on-topic but vague	understandable, minor errors

Score	Empathy	Relevance	Fluency
5	reflects how the user feels *now*, including any change	responds to the specifics of the latest turn	natural and error-free

Pairwise task: "Which reply better responds to how the speaker feels now?" (A / B / tie). System names are hidden, A/B order and item order are randomised per rater, and ties are excluded from the sign test. Agreement on 1–5 ratings uses ordinal Krippendorff's α.

Appendix D. Week-by-week plan

Dates	Milestone	Lead role
5–9 Oct	data loaders for ED, ESConv, DailyDialog and MELD; recogniser trained	Recogniser, Data
12 Oct	**checkpoint:** recogniser F1 and Δ statistics; τ fixed	Data
12–16 Oct	B0 checkpoint; evaluation scripts (perplexity, Distinct-n, judge)	Generator, Auto eval
19–23 Oct	B1 and ETG trained	Generator
26–30 Oct	ETG-noΔ and ETG-noGate; second seeds; tuning	Generator, Auto eval
2–6 Nov	automatic metrics, swap test, gate analysis; human evaluation	Auto eval, Human eval
10 Nov	**presentation** (≥90% of results)	All
11–20 Nov	error analysis, qualitative cases, code clean-up	All
27 Nov	**report, code and data submitted**	All